

IR Generation

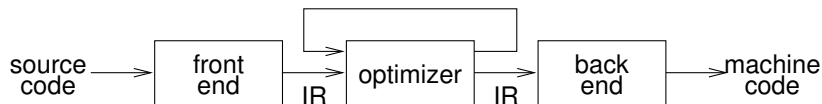
Intermediate representations

Why use an intermediate representation?

- ① break the compiler into manageable pieces
 - good software engineering technique
- ② simplifies retargeting to new host
 - isolates back end from front end
- ③ simplifies handling of “poly-architecture” problem
 - m lang's, n targets $\Rightarrow m + n$ components
- ④ enables machine-independent optimization
 - general techniques, multiple passes

An intermediate representation is a compile-time data structure

Intermediate representations



Generally speaking:

- front end produces IR
- optimizer transforms that representation into an equivalent program that may run more efficiently
- back end transforms IR into native code for the target machine

Intermediate representations

Representations talked about in the literature include:

- abstract syntax trees (AST)
- linear (operator) form of tree
- directed acyclic graphs (DAG)
- control flow graphs
- program dependence graphs
- static single assignment form
- 3-address code
- hybrid combinations

Intermediate representations - properties

Important IR Properties

- ease of generation
- ease of manipulation
- cost of manipulation
- level of abstraction
- size of typical procedure

Subtle design decisions in the IR have far reaching effects on the speed and effectiveness of the compiler.

Level of exposed detail is a crucial consideration.

IR design issues

- Is the chosen IR appropriate for the (analysis/ optimization/ transformation) passes under consideration?
- What is the IR level: close to language/machine.
- Multiple IRs in a compiler: for example, High, Medium and Low

t1 $\leftarrow$ a[i,j+2]

t1 $\leftarrow$ j + 2

r1 $\leftarrow$ [fp-4]

t2 $\leftarrow$ i * 20

r2 $\leftarrow$ r1 + 2

t3 $\leftarrow$ t1 + t2

r3 $\leftarrow$ [fp-8]

t4 $\leftarrow$ 4 * t3

r4 $\leftarrow$ r3 * 20

t5 $\leftarrow$ addr a

r5 $\leftarrow$ r4 + r2

t6 $\leftarrow$ t5 + t4

r6 $\leftarrow$ 4 * r5

t7 $\leftarrow$ *t6

r7 $\leftarrow$ fp - 216

f1 $\leftarrow$ [r7+r6]

(a)

(b)

(c)

(a) High-, (b) medium-, and (c) low-level representations of a C array reference.

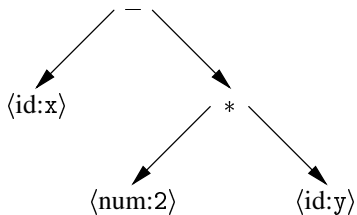
Intermediate representations

Broadly speaking, IRs fall into three categories:

- Structural
 - structural IRs are graphically oriented
 - examples include trees, DAGs
 - heavily used in source to source translators
 - nodes, edges tend to be large
- Linear
 - pseudo-code for some abstract machine
 - large variation in level of abstraction
 - simple, compact data structures
 - easier to rearrange
- Hybrids
 - combination of graphs and linear code
 - attempt to take best of each
 - e.g., control-flow graphs
 - Example: GCC Tree IR.

Abstract syntax tree

An abstract syntax tree (AST) is just the parse tree with the nodes for most non-terminal symbols removed.



This represents "x - 2 * y".

SDD for generating AST: Example-1

$E \rightarrow E + T$
$ E - T$
$ T$
$T \rightarrow T * F$
$ F$
$F \rightarrow \text{id}$

SDD for generating AST: Example-1

$E_1 \rightarrow E_2 + T$	$E_1.node = \text{new Node}(+, E_2.node, T.node)$
$\quad E_2 - T$	$E_1.node = \text{new Node}(-, E_2.node, T.node)$
$\quad T$	$E_1.node = T.node$
$T_1 \rightarrow T_2 * F$	$T_1.node = \text{new Node}(*, T_2.node, F.node)$
$\quad F$	$T_1.node = F.node$
$F \rightarrow \text{id}$	$F.node = \text{new Leaf}(\text{id.lexeme})$

SDD for generating AST: Example-2

$$\begin{array}{c|c} \hline \hline E \rightarrow TE' & \\ E'_1 \rightarrow +TE'_2 & \\ & | \epsilon \\ T \rightarrow FT' & \\ T'_1 \rightarrow *FT'_2 & \\ & | \epsilon \\ F \rightarrow \text{id} & \\ \hline \hline \end{array}$$

SDD for generating AST: Example-2

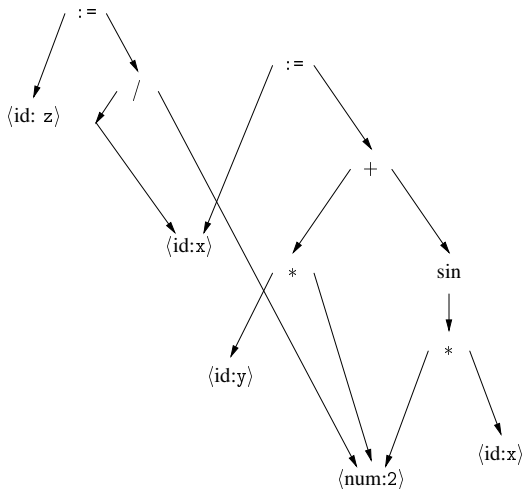
$E \rightarrow TE'$	$E.node = E'.node$ $E'.in = T.node$
$E'_1 \rightarrow +TE'_2$	$E'_1.node = E'_2.node$ $E'_2.in = \text{new Node}(+, E'_1.in, T.node)$
$\quad \mid \epsilon$	$E'_1.node = E'_1.in$
$T \rightarrow FT'$	$T.node = T'.node$ $T'.in = F.node$
$T'_1 \rightarrow *FT'_2$	$T'_1.node = T'_2.node$ $T'_2.in = \text{new Node}(*, T'_1.in, F.node)$
$\quad \mid \epsilon$	$T'_1.node = T'_1.in$
$F \rightarrow \text{id}$	$F.node = \text{new Leaf}(\text{id.lexeme})$

Homework: Draw the dependency graph for $x-2*y$

Directed acyclic graph

A directed acyclic graph (DAG) is an AST with a unique node for each value.

```
x := 2 * y + sin(2*x)
z := x / 2
```



Value-number methods for constructing DAGs:

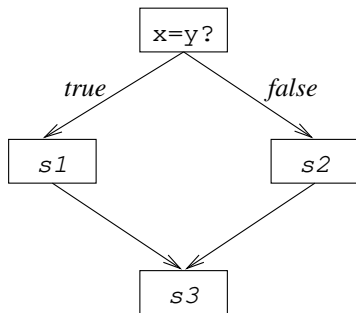
- (1) Assign a unique number to every node.
- (2) Search for a node with signature $(op, valLeft, valRight)$.

Control flow graph

The control flow graph (CFG) models the transfers of control in the procedure

- nodes in the graph are basic blocks
straight-line blocks of code
- edges in the graph represent control flow
loops, if-then-else, case, goto

```
if (x=y) then  
    s1  
else  
    s2  
s3
```



3-address code

- At most one operator on the right side of an instruction.
- 3-address code can mean a variety of representations.
- In general, it allows statements of the form:

$$x \leftarrow y \text{ op } z$$

with a single operator and, at most, three names.

Simpler form of expression:

$$x = 2 * y$$

becomes

$$t1 \leftarrow 2 * y$$

$$t2 \leftarrow x - t1$$

Advantages

- compact form (direct naming)
- names for intermediate values

3-address code: Addresses

Three-address code is built from two concepts: addresses and instructions.

- An address can be
 - A name: source variable program name or pointer to the Symbol Table name.
 - A constant: Constants in the program.
 - Compiler generated temporary.

3-address code

Typical instructions types include:

- ① assignments $x \leftarrow y \text{ op } z$
- ② assignments $x \leftarrow \text{op } y$
- ③ assignments $x \leftarrow y[i]$
- ④ assignments $x \leftarrow y$
- ⑤ branches `goto L`
- ⑥ conditional branches
`if x goto L`
- ⑦ procedure calls
`param x1, param x2, ... param xn`
`and`
`call p, n`
- ⑧ address and pointer assignments

How to translate:

```
if (x < y) S1 else  
S2
```

?

3-address code - implementation

Quadruples

- Has four fields: op, arg1, arg2 and result.
- Some instructions (e.g. unary minus) do not use arg2.
- For copy statement : the operator itself is =; for others it is implied.
- Instructions like `param` don't use neither arg2 nor result.
- Jumps put the target label in result.

$x - 2 * y$

	<u>op</u>	<u>result</u>	<u>arg1</u>	<u>arg2</u>
(1)	load	t1	y	
(2)	loadi	t2	2	
(3)	mult	t3	t2	t1
(4)	load	t4	x	
(5)	sub	t5	t4	t3

- simple record structure with four fields
- easy to reorder
- explicit names

3-address code - implementation

Triples

$x - 2 * y$			
(1)	load	y	
(2)	loadi	2	
(3)	mult	(1)	(2)
(4)	load	x	
(5)	sub	(4)	(3)

- use table index as implicit name
- require only three fields in record
- harder to reorder

3-address code - implementation

Indirect Triples

$$x - 2 * y$$

	exec-order	stmt	op	arg1	arg2
(1)	(100)	(100)	load	y	
(2)	(101)	(101)	loadi	2	
(3)	(102)	(102)	mult	(100)	(101)
(4)	(103)	(103)	load	x	
(5)	(104)	(104)	sub	(103)	(102)

- simplifies moving statements (change the execution order)
- more space than triples
- implicit name space management

Indirect triples advantage

```
for i:=1 to 10 do
begin
  a=b*c
  d=i*3
end
```

(a)

Optimized version

```
a=b*c
for i:=1 to 10 do
begin
  d=i*3
end
```

(b)

```
(1) := 1 i
(2) nop
(3) * b c
(4) := (3) a
(5) * 3 i
(6) := (5) d
(7) + 1 i
(8) := (7) i
(9) LE i 10
(10) IFT goto (2)
```

Execution Order (a) : 1 2 3 4 5 6 7
8 9 10

Execution Order (b) : 3 4 1 2 5 6 7
8 9 10

Other hybrids

An attempt to get the best of both worlds.

- graphs where they work
- linear codes where it pays off

Unfortunately, there appears to be little agreement about where to use each kind of IR to best advantage.

A commonly used hybrid IR is a control flow graph with low-level, three address code for each basic block.

Intermediate representations

But, this isn't the whole story

Symbol table:

- identifiers, procedures
- size, type, location
- lexical nesting depth

Constant table:

- representation, type
- storage class, offset(s)

Storage map:

- storage layout
- overlap information
- (virtual) register assignments

Gap between HLL and IR

Gap between HLL and IR

- High level languages may allow complexities that are not allowed in IR (such as expressions with multiple operators).
- High level languages have many syntactic constructs, not present in the IR (such as if-then-else or loops)

Challenges in translation:

- Deep nesting of constructs.
- Recursive grammars.
- We need a systematic approach to IR generation.

Goal:

- A HLL to IR translator.
- Input: A program in HLL.
- Output: A program in IR (may be an AST or program text)

Translating expressions

```
S -> id = E;    {S.code = E.code ||
                  gen(top.get(id.lexeme) '=' E.addr);}

E -> E1 + E2     {E.addr = new Temp();
                  E.code = E1.code || E2.code ||
                  gen(E.addr '=' E1.addr '+' E2.addr);}

| - E1          {E.addr = new Temp();
                  E.code = E1.code ||
                  gen(E.addr '=' - E2.addr);}

| (E1)          {E.addr = E1.addr;
                  E.code = E1.code}

| id            {E.addr = top.get(id.lexeme);
                  E.code = ' ' }
```

- addr: a synthesized-attribute of E – denotes the address holding the val of E .
- top is the top-most (current) symbol table.

Translating expressions: Incremental Translation

$S \rightarrow id = E; \quad \{ \text{gen}(\text{top.get}(id.\text{lexeme}) \text{ '=' } E.\text{addr}); \}$

$E \rightarrow E1 + E2 \quad \{ E.\text{addr} = \text{new Temp}();$
 $\quad \text{gen}(E.\text{addr} \text{ '=' } E1.\text{addr} \text{ '+' } E2.\text{addr}); \}$

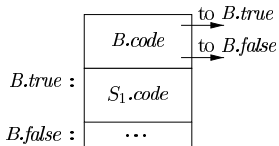
$\quad | - E1 \quad \{ E.\text{addr} = \text{new Temp}();$
 $\quad \text{gen}(E.\text{addr} \text{ '=' } - E2.\text{addr}); \}$

$\quad | (E1) \quad \{ E.\text{addr} = E1.\text{addr}; \}$

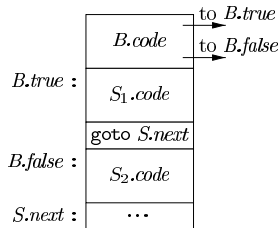
$\quad | id \quad \{ E.\text{addr} = \text{top.get}(id.\text{lexeme}); \}$

- Constructs a three-address instruction and appends the instruction to the sequence of instructions.

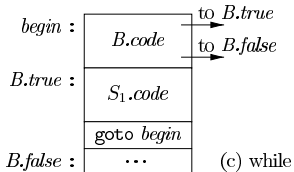
IR generation for flow-of-control statements

$$S \rightarrow \text{if } (B) S_1$$
$$S \rightarrow \text{if } (B) S_1 \text{ else } S_2$$
$$S \rightarrow \text{while } (B) S_1$$


(a) if



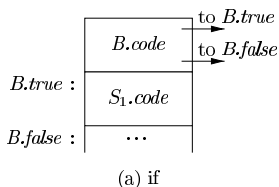
(b) if-else



(c) while

IR generation for `if`-statement

code	Code for the node; <u>synthesized</u>
next	Label of the next instruction; <u>inherited</u>
true	Label of next instruction if boolean expression evaluates to <i>true</i> ; <u>inherited</u>
false	Label of next instruction if boolean expression evaluates to <i>false</i> ; <u>inherited</u>



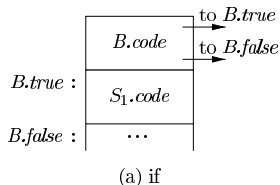
```
P->S      {S.next = new Label();  
           P.code = S.code || label(S.next)}
```

```
S->assign  {S.code = assign.code}
```

```
S->if (B) S1
```

IR generation for if-statement

code	Code for the node; <u>synthesized</u>
next	Label of the next instruction; <u>inherited</u>
true	Label of next instruction if boolean expression evaluates to <i>true</i> ; <u>inherited</u>
false	Label of next instruction if boolean expression evaluates to <i>false</i> ; <u>inherited</u>



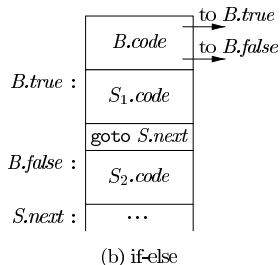
```
P->S      {S.next = new Label();  
           P.code = S.code || label(S.next)}
```

```
S->assign  {S.code = assign.code}
```

```
S->if (B) S1 {B.true = new Label();  
            B.false = S1.next = S.next  
            S.code = B.code || label(B.true) || S1.code}
```

IR generation for if else-statement

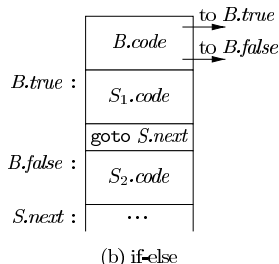
code	Code for the node; <u>synthesized</u>
next	Label of the next instruction; <u>inherited</u>
true	Label of next instruction if boolean expression evaluates to <i>true</i> ; <u>inherited</u>
false	Label of next instruction if boolean expression evaluates to <i>false</i> ; <u>inherited</u>



```
S->if (B) S1
    else S2
```

IR generation for if else-statement

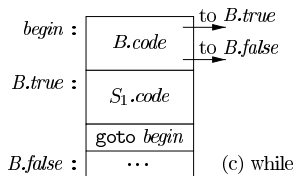
code	Code for the node; <u>synthesized</u>
next	Label of the next instruction; <u>inherited</u>
true	Label of next instruction if boolean expression evaluates to <i>true</i> ; <u>inherited</u>
false	Label of next instruction if boolean expression evaluates to <i>false</i> ; <u>inherited</u>



```
S->if (B) S1    {B.true = new Label();
    else S2     B.false = new Label();
                S1.next = S2.next = S.next;
                S.code = B.code || label(B.true) || S1.code
                        || gen ('goto' S.next)
                        || label (B.false) || S2.code}
```

IR generation for `while`-statement

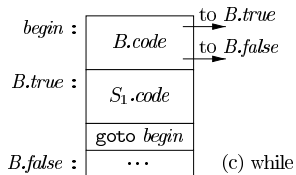
code	Code for the node; <u>synthesized</u>
next	Label of the next instruction; <u>inherited</u>
true	Label of next instruction if boolean expression evaluates to <i>true</i> ; <u>inherited</u>
false	Label of next instruction if boolean expression evaluates to <i>false</i> ; <u>inherited</u>



`S->while(B) S1`

IR generation for `while`-statement

code	Code for the node; <u>synthesized</u>
next	Label of the next instruction; <u>inherited</u>
true	Label of next instruction if boolean expression evaluates to <i>true</i> ; <u>inherited</u>
false	Label of next instruction if boolean expression evaluates to <i>false</i> ; <u>inherited</u>



```
S->while(B) S1    {begin = new Label();
                   B.true = new Label();
                   B.false = S.next
                   S1.next = begin
                   S.code = label(begin) || B.code
                           || label(B.true) || S1.code
                           || gen('goto' begin) }
```

IR generation for boolean expressions

$$B \rightarrow B \parallel B \mid B \&\& B \mid !B \mid (B) \\ \mid E \text{ rel } E \mid \text{true} \mid \text{false}$$

- For $B \rightarrow B_1 \parallel B_2$, if B_1 evaluates to **true**, then B_2 should not be evaluated, and control should transfer to $B.\text{true}$.
 - If B_1 evaluates to **false**, then B_2 should be evaluated.
- For $B \rightarrow B_1 \&\& B_2$, if B_1 evaluates to **false**, then B_2 should not be evaluated, and control should transfer to $B.\text{false}$,
 - If B_1 evaluates to **true**, then B_2 should be evaluated.
- **rel** corresponds to comparison operators: $<, \leq, ==, !=, >, \geq$.

IR generation for boolean expressions

$B \rightarrow E1 \text{ rel } E2$

$B \rightarrow \text{true}$

$B \rightarrow \text{false}$

IR generation for boolean expressions

```
B -> E1 rel E2      t = new Temp()
                     B.code=E1.code||E2.code
                     || gen(t=='E1.addr rel.op E2.addr)
                     || gen('if' t 'goto' B.true)
                     || gen('goto' B.false);

B -> true             B.code = gen('goto' B.true)

B -> false            B.code = gen('goto' B.false)
```

IR generation for boolean expressions

$B \rightarrow B1 \mid\mid B2$

$B \rightarrow B1 \&\& B2$

$B \rightarrow !B1$

IR generation for boolean expressions

B -> B1 B2	B1.true = B.true B1.false = new Label() B2.true = B.true B2.false = B.false B.code = B1.code label(B1.false) B2.code
B -> B1 && B2	B1.true = new Label() B1.false = B.false B2.true = B.true B2.false = B.false B.code = B1.code label(B1.true) B2.code
B -> !B1	B1.true = B.false B1.false = B.true B.code = B1.code

IR generation for control-flow statements: Example

HLL Code:

```
if ( x < 100 || x > 200 && x != y ) x = 0;
```

IR Code:

```
    if x < 100 goto L1
    goto L3
L3:  if x > 200 goto L4
    goto L2
L4:  if x != y goto L1
    goto L2
L1:  x = 0
L2:
```

An optimization to avoid redundant `gotos`

- Consider the HLL code: `if (x > 100) x = 0;`
- Using the previous SDD, we get the following IR:

```
    if x > 100 goto L1
    goto L2
```

```
L1: x = 0
```

```
L2:
```

- But this is equivalent to :

```
    ifFalse x > 100 goto L2
    x = 0
```

```
L2:
```

- Similarly, while translating `B1 || B2`, since code for `B2` immediately follows code for `B1`, there is no need of an unconditional jump for `B1.false`.
- To avoid redundant `gotos`, we can use a special `fall` label.
 - It indicates that the instruction to be executed immediately follows.

An optimization to avoid redundant `gotos`

```
S->if (B) S1      {B.true = fall;  
                  B.false = S1.next = S.next  
                  S.code = B.code || S1.code}  
  
B->E1 rel E2
```

An optimization to avoid redundant `gotos`

`S->if (B) S1`

```
{B.true = fall;  
B.false = S1.next = S.next  
S.code = B.code || S1.code}
```

`B->E1 rel E2`

```
{t = new Temp()  
B.code=E1.code||E2.code  
|| gen(t=='E1.addr rel.op E2.addr)  
|| if (B.true=fall)  
    gen('ifFalse' t 'goto' B.false)  
else if (B.false=fall)  
    gen('if' t 'goto' B.true)  
else  
    gen ('if' t 'goto' B.true)  
|| gen('goto' B.false) }
```

Homework: Use `fall` to optimize IRs for `B1 || B2` and `B1 && B2`

One-Pass Code generation using Backpatching

- While generating code for `if (B) S`, code for `B` is generated before code for `S`.
- However, code for `B` contains a jump for instruction after `S`.
 - The target address for this jump can therefore only be known after generating code for `S`.
 - This requires a second pass to fill up the jump addresses.
- In Backpatching, instead of having jump targets as inherited attributes of boolean expressions, jump instructions with holes become synthesized attributes of branch statements.
 - Self-reading exercise: Chapter 6, Section 6.7
 - **Part of the syllabus.**

Array elements dereference (Recall)

- Elements are typically stored in a block of consecutive locations.
- If the width of each array element is w , then the i^{th} element of array A (say, starting at the address $base$), begins at the location:

$$base + i \times w$$

- We declare arrays by the number of elements (n_j is the size of the j^{th} dimension) and the width of each element in an array is fixed (say w). The location for $A[i_1][i_2]$ is given by

$$base + (i_1 \times n_2 + i_2) \times w$$

- For multi-dimensions, beginning address of $A[i_1][i_2]$ is calculated by the formula:

$$base + i_1 \times w_1 + i_2 \times w_2$$

where, w_1 is the width of the row, and w_2 is the width of one element.

Array elements dereference (Recall)

- If the width of each array element is w , then the i^{th} element of array A (say, starting at the address $base$), begins at the location:

$$base + i \times w$$

- **Question:** If the array index does not start at '0', at some 'k' then, how to calculate the address of $A[i]$?
- **Homework:** What if the data is stored in column-major form?

Translation of Array references

- Extending the expression grammar with arrays:

$S \rightarrow id = E;$

$\quad | L = E;$

$E \rightarrow E1 + E2$

$\quad | id$

$\quad | L$

$L \rightarrow id [E]$

$\quad | L1 [E]$

Translation of Array references (cont.)

```
S -> id = E; {gen(top.get(id.lexeme) '=' E.addr) }

      | L = E; {gen(L.array.base['L.addr'] '=' E.addr); }

E -> E1 + E2 {E.addr = new Temp();
              gen(E.addr '=' E1.addr '+' E2.addr); }

      | id      {E.addr = top.get(id.lexeme); }

      | L      {E.addr = new Temp();
                 gen(E.addr '=' L.array.base['L.addr']); }
```

Nonterminal *L* has three synthesized attributes

- 1 *L.addr* denotes a temporary that is used while computing the offset for the array reference.
- 2 *L.array* is a pointer to the ST entry for the array name. The field *base* gives the actual l-value of the array reference.

Translation of Array references (contd)

Type and width:

- Let a denotes a 2×3 integer array.
- Type of a is given by $\text{array}(2, \text{array}(3, \text{integer}))$
- Width of $a = 24$ (size of $\text{integer} = 4$).
- Type of $a[i]$ is $\text{array}(3, \text{integer})$, width = 12.
- Type of $a[i][j] = \text{integer}$

Translation of Array references (contd)

```
L -> Id [E]  {L.array = top.get(id.lexeme);  
               L.type = L.array.type.elem;  
               L.addr = new Temp();  
               gen(L.addr '=' E.addr '*' L.type.width);}
```

```
| L1 [E]  {L.array = L1.array;  
           L.type = L1.type.elem;  
           t = new Temp();  
           L.addr = new Temp();  
           gen(t '=' E.addr '*' L.type.width);  
           gen (L.addr '=' L1.addr '+' t);}
```

③ *L.type* is the type of the subarray generated by *L*.

- For any type *t*: *t.width* gives get the width of the type.
- For any type *t*: *t.elem* gives the element type.

Translation of Array references (contd)

Example:

- Let a denotes a 2×3 integer array.
- Type of a is given by $\text{array}(2, \text{array}(3, \text{integer}))$
- Width of $a = 24$ (size of $\text{integer} = 4$).
- Type of $a[i]$ is $\text{array}(3, \text{integer})$, width = 12.
- Type of $a[i][j] = \text{integer}$

Exercise:

- Write three address code for $c + a[i][j]$

```
t1 = i * 12
t2 = j * 4
t3 = t1 + t2
t4 = a [t3]
t5 = c + t4
```

Some challenges/questions

- How to translate implicit branches: `break` and `continue`?
- How to translate `switch` statements efficiently?
- How to translate procedure code?

Self-reading exercise: Dragon Book, Chapter 6, Sections 6.7,6.8,6.9